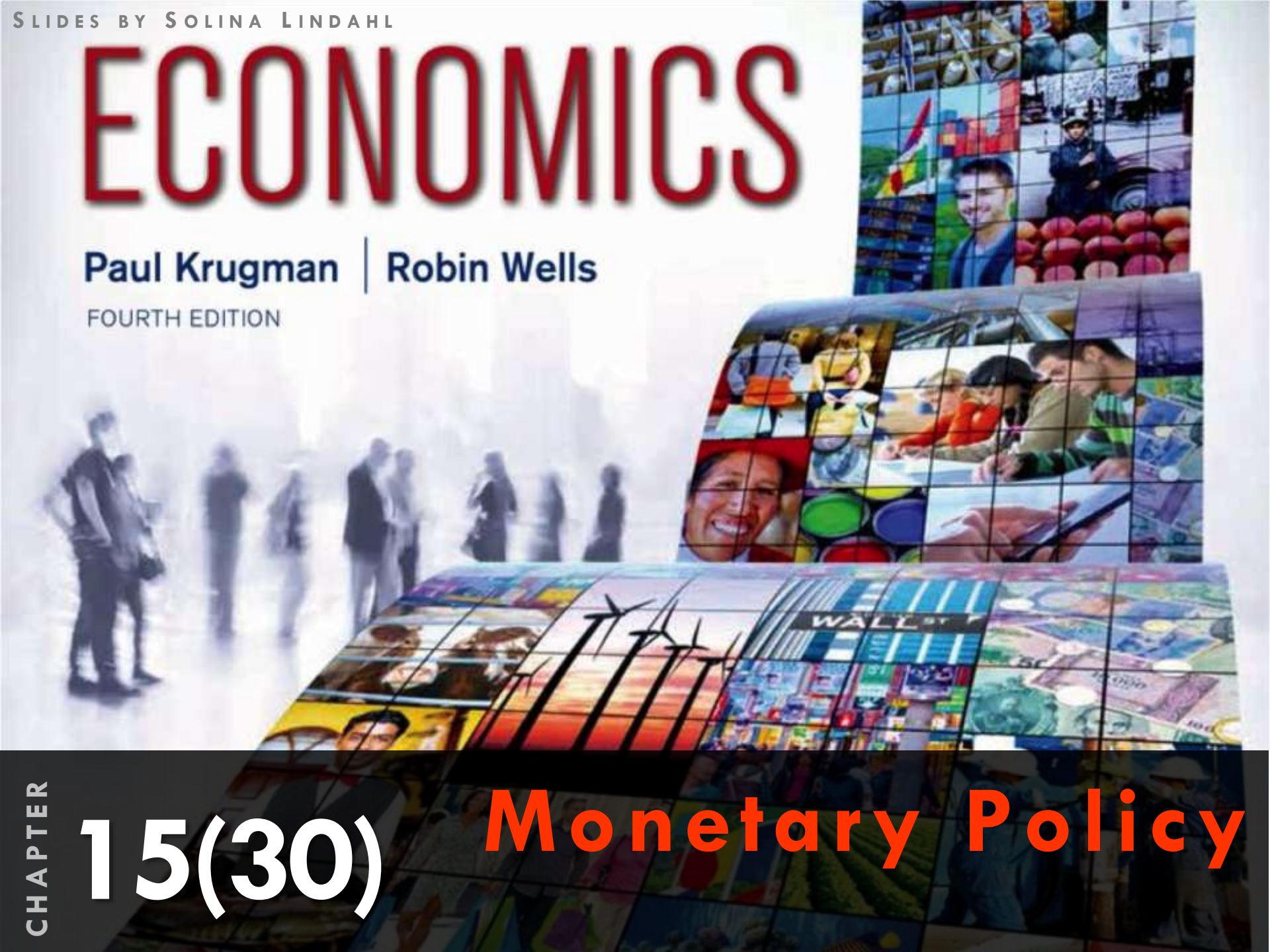


# ECONOMICS

Paul Krugman | Robin Wells

FOURTH EDITION



CHAPTER

15(30)

Monetary Policy

# ► What you will learn in this chapter



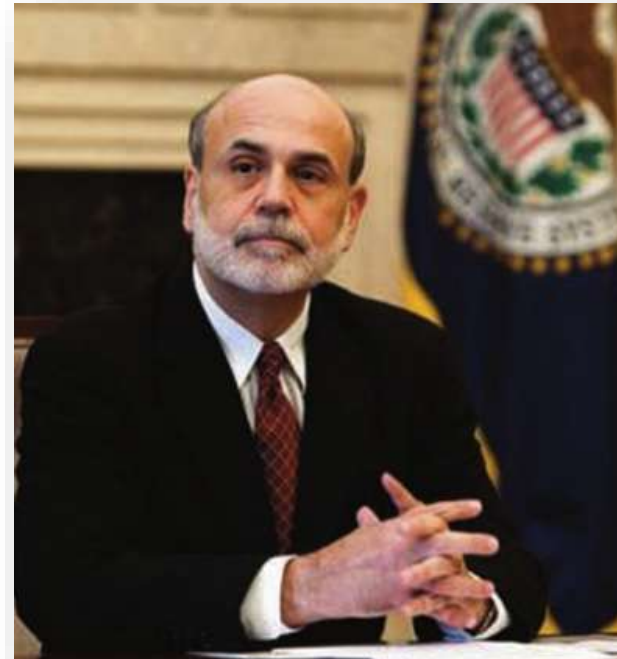
- What the money demand curve is
- Why the liquidity preference model determines the interest rate in the short run
- How the Federal Reserve implements monetary policy, moving the interest rate to affect aggregate output
- Why monetary policy is the main tool for stabilizing the economy
- How the behavior of the Federal Reserve compares to that of other central banks
- Why economists believe in monetary neutrality—that monetary policy affects only the price level, not aggregate output, in the long run

# THE MOST POWERFUL PERSON IN GOVERNMENT

**Janet Yellen, Current Chair**



**Ben Bernanke, Former Chair**



**The chairmanship of the Federal Reserve is arguably the most powerful position in the U.S. government**

# THE DEMAND FOR MONEY



*There is a price to be paid for the convenience of holding money.*

## The Opportunity Cost of Holding Money

**We all carry some cash around for the convenience.**

**When we do, we give up interest income we'd collect if that spending power were in an interest-bearing asset like a bond.**

# INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

**Individuals and firms trade off the benefit of holding cash versus the benefit of holding interest-bearing nonmonetary assets.**

**Selected interest rates, 2007 (before the financial crisis)**

|                                         |       |
|-----------------------------------------|-------|
| One-month certificates of deposit (CDs) | 5.30% |
| Interest-bearing demand deposits        | 2.30% |
| Currency                                | 0     |

*Source: Federal Reserve Bank of St. Louis*

# INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

***Short-term interest rates:*** the interest rates on financial assets that mature within six months or less.

***Long-term interest rates:*** interest rates on financial assets that mature a number of years in the future.

The ***higher*** the interest rate, the ***higher*** the opportunity cost of holding money.

The ***lower*** the interest rate, the ***lower*** the opportunity cost of holding money.

# INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Although there are many different interest rates and they don't move exactly together, as a rule **most short-term rates tend to move in the same direction.**

|                                                                       | June 2007   | June 2008   |
|-----------------------------------------------------------------------|-------------|-------------|
| Federal funds rate                                                    | 5.25%       | 2.00%       |
| One-month certificates of deposit (CDs)                               | 5.30%       | 2.50%       |
| Interest-bearing demand deposits                                      | 2.30%       | 1.24%       |
| Currency                                                              | 0           | 0           |
| <b>CDs minus interest-bearing demand deposits (percentage points)</b> | <b>3.00</b> | <b>1.26</b> |
| <b>CDs minus currency (percentage points)</b>                         | <b>5.30</b> | <b>2.50</b> |

Source: Federal Reserve Bank of St. Louis

# THE MONEY DEMAND CURVE

Interest rate,  $r$

*At higher interest rates, the cost of holding money is greater, so less quantity is demanded.*

The **money demand curve** shows the relationship between the quantity of money demanded and the interest rate.

*At lower interest rates, the cost of holding money is smaller, so a higher quantity is demanded.*

Money demand curve, MD

Quantity of money

# SHIFTS OF THE REAL MONEY DEMAND CURVE

**The following factors shift the money demand curve:**

## **Changes in aggregate price level**

*Higher prices mean we need more money for transactions (and vice versa).*

## **Changes in real GDP**

*More goods and services produced and sold means we need more money (and vice versa).*

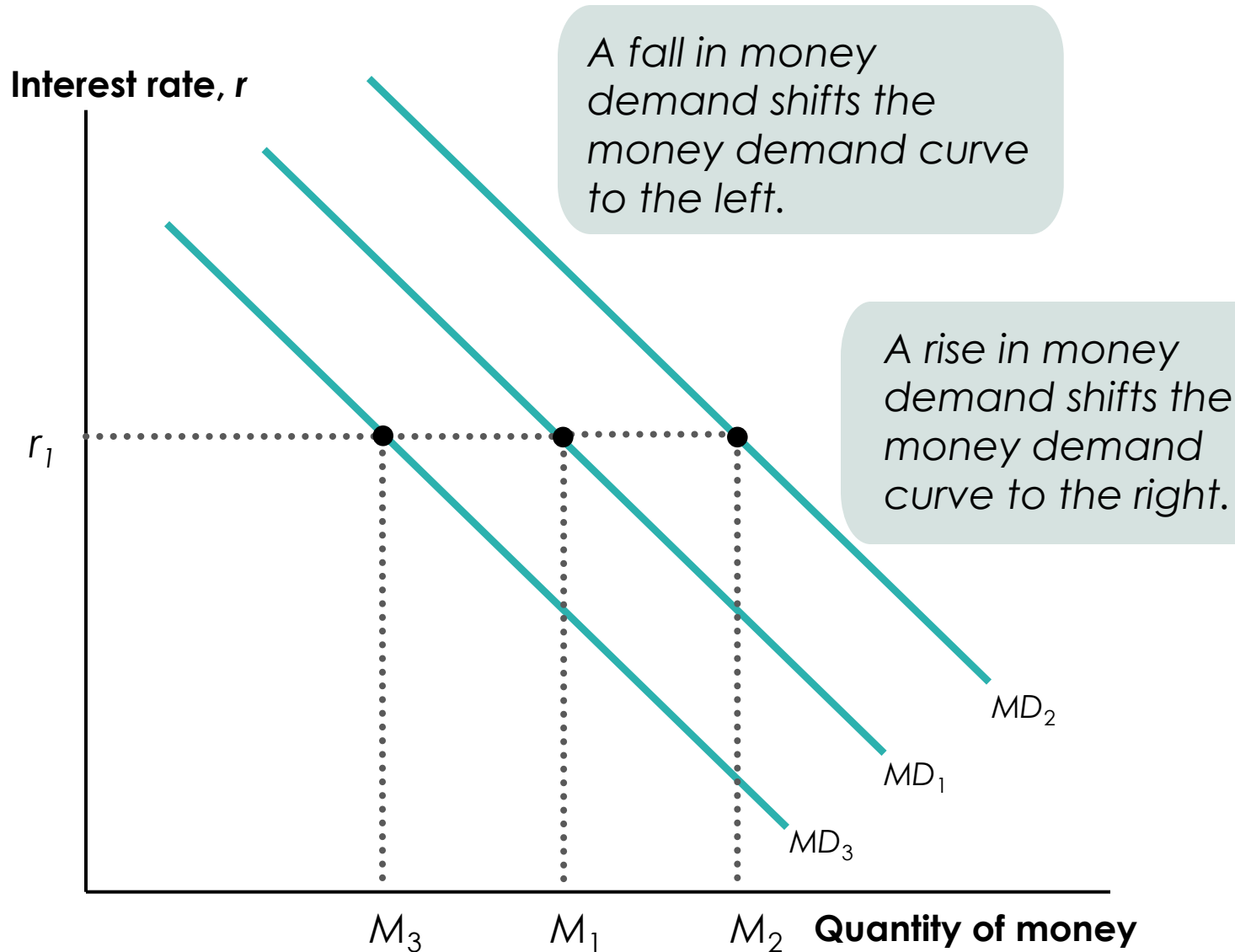
## **Changes in technology**

*The ease of credit card payments reduces the need for cash.*

## **Changes in institutions**

*After 1980 banks were allowed to offer interest on checking accounts. This decreased the cost of holding money, and money demand increased.*

# SHIFTS IN THE DEMAND FOR MONEY



# LEARN BY DOING: DISCUSS

**With a partner, fill in the table.**

| Situation                                                                                                  | Shift in the demand for money (provide direction of shift) | Shift in the supply of money (provide direction of shift) | Effect on the equilibrium interest rate | Effect on the equilibrium quantity of money |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------|---------------------------------------------|
| The information technology for making transactions improves                                                |                                                            |                                                           |                                         |                                             |
| The Federal Reserve makes an open-market purchase of Treasury bills                                        |                                                            |                                                           |                                         |                                             |
| Real GDP increases in the economy                                                                          |                                                            |                                                           |                                         |                                             |
| Real GDP decreases in the economy, and the Federal Reserve makes an open-market purchase of Treasury bills |                                                            |                                                           |                                         |                                             |
| The aggregate price level increases                                                                        |                                                            |                                                           |                                         |                                             |

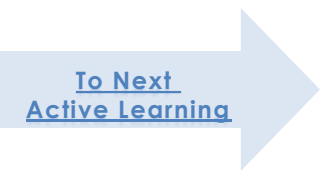
[To answer](#)

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# LEARN BY

(Answers to previous slide)

| Situation                                                                                                  | Shift in the demand for money (provide direction of shift)                                     | Shift in the supply of money (provide direction of shift)                                                    | Effect on the equilibrium interest rate     | Effect on the equilibrium quantity of money         |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|
| The information technology for making transactions improves                                                | Demand for money shifts left, since at every interest rate people will wish to hold less money | There is a movement along the supply of money curve due to the shift in the demand for money                 | The equilibrium interest rate will decrease | The equilibrium quantity of money will be unchanged |
| The Federal Reserve makes an open-market purchase of Treasury bills                                        | There is a movement along the demand for money curve due to the shift in the supply of money   | Supply of money will shift to the right, as the Federal Reserve injects new reserves into the banking system | The equilibrium interest rate will decrease | The equilibrium quantity of money will increase     |
| Real GDP increases in the economy                                                                          | Demand for money will shift to the right when real GDP increases                               | There is a movement along the supply of money curve due to the shift in the demand for money                 | The equilibrium interest rate will increase | The equilibrium quantity of money will be unchanged |
| Real GDP decreases in the economy, and the Federal Reserve makes an open-market purchase of Treasury bills | Demand for money will shift to the left when real GDP decreases                                | Supply of money will shift to the right when the Federal Reserve makes an open-market purchase               | The equilibrium interest rate will decrease | The equilibrium quantity of money will increase     |
| The aggregate price level increases                                                                        | Demand for money shifts to the right when the aggregate price level increases                  | There will be a movement along the supply of money curve due to the shift in the demand for money            | The equilibrium interest rate will increase | The equilibrium quantity of money will be unchanged |



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# MONEY AND INTEREST RATES

**So how does the Fed control interest rates?**

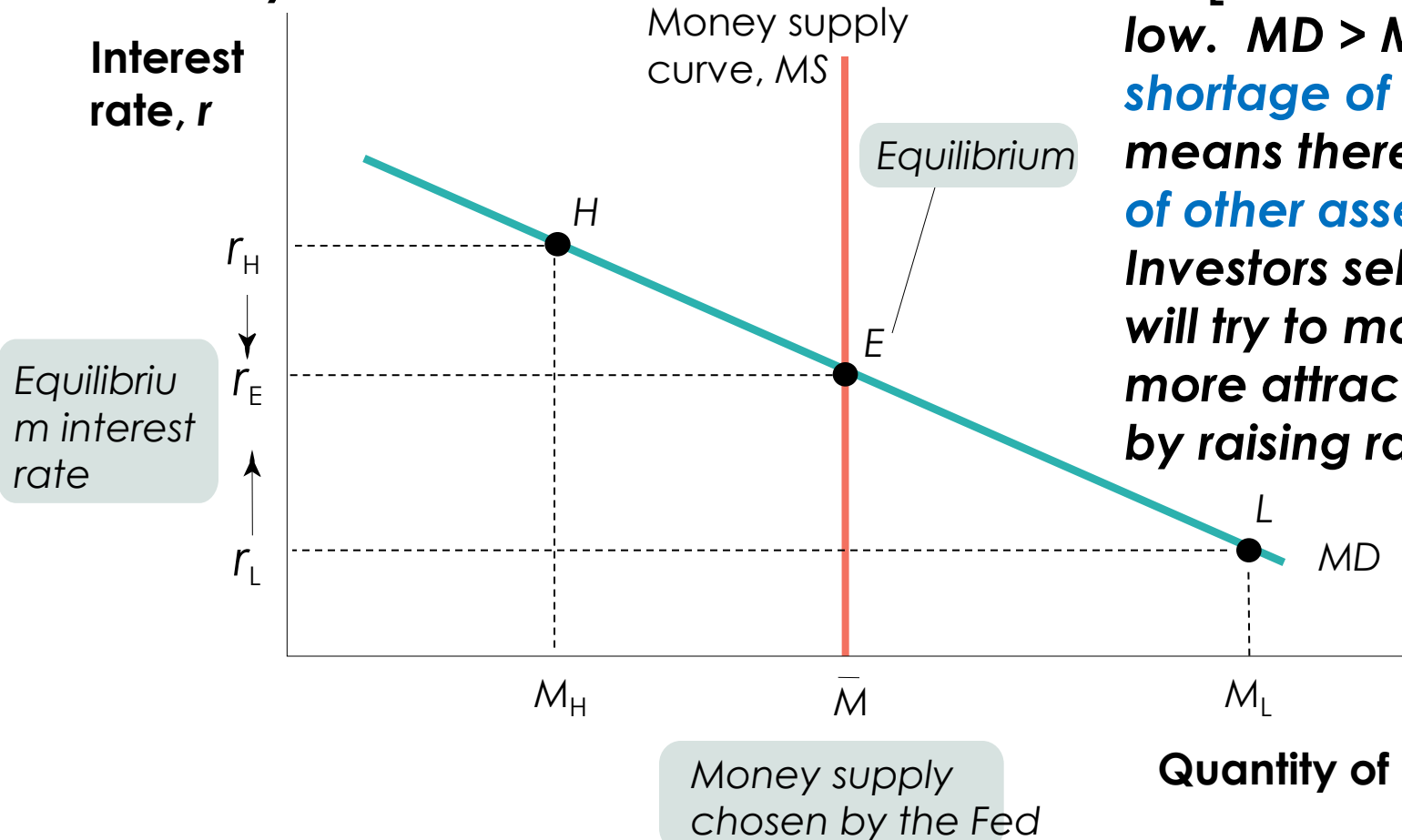
**We need to understand how interest rates are set in the market.**

***The liquidity preference model of the interest rate*** asserts that the interest rate is determined by the supply and demand for money.

***The money supply curve*** shows how the nominal quantity of money supplied varies with the interest rate.

# EQUILIBRIUM IN THE MONEY MARKET

At  $r_H$ , interest rates are high.  $MS > MD$ , and a **surplus of money** means there is a **shortage of other assets** like bonds. Investors selling bonds will realize they can reduce interest rates and still find buyers.



At  $r_L$ , interest rates are low.  $MD > MS$ , and a **shortage of money** means there is a **surplus of other assets** like bonds. Investors selling bonds will try to make them more attractive to buyers by raising rates.

## LEARN BY DOING: PRACTICE QUESTION

**Suppose that an economy's aggregate output level increases. Assuming that the central bank does not engage in monetary policy, which of the following statements is true?**

- a) The interest rate will increase.**
- b) The interest rate will decrease.**
- c) The interest rate will be unaffected by this change in aggregate output.**
- d) This change in aggregate output will cause a movement along the money demand curve.**

# MONETARY POLICY AND THE INTEREST RATE

By adjusting the money supply up or down, the Fed can set the interest rate.

The Fed sets a **target federal funds rate** and **undertakes the appropriate open-market operations**.

***Target federal funds rate:*** the Federal Reserve's desired federal funds rate.

# MONETARY POLICY AND THE INTEREST RATE

## The Effect of an Increase in the Money Supply on the Interest Rate

Interest  
rate,  $r$

An increase  
in the money  
supply ...

$MS_1$

$MS_2$

$r_1$

$E_1$

$r_2$

$E_2$

$MD$

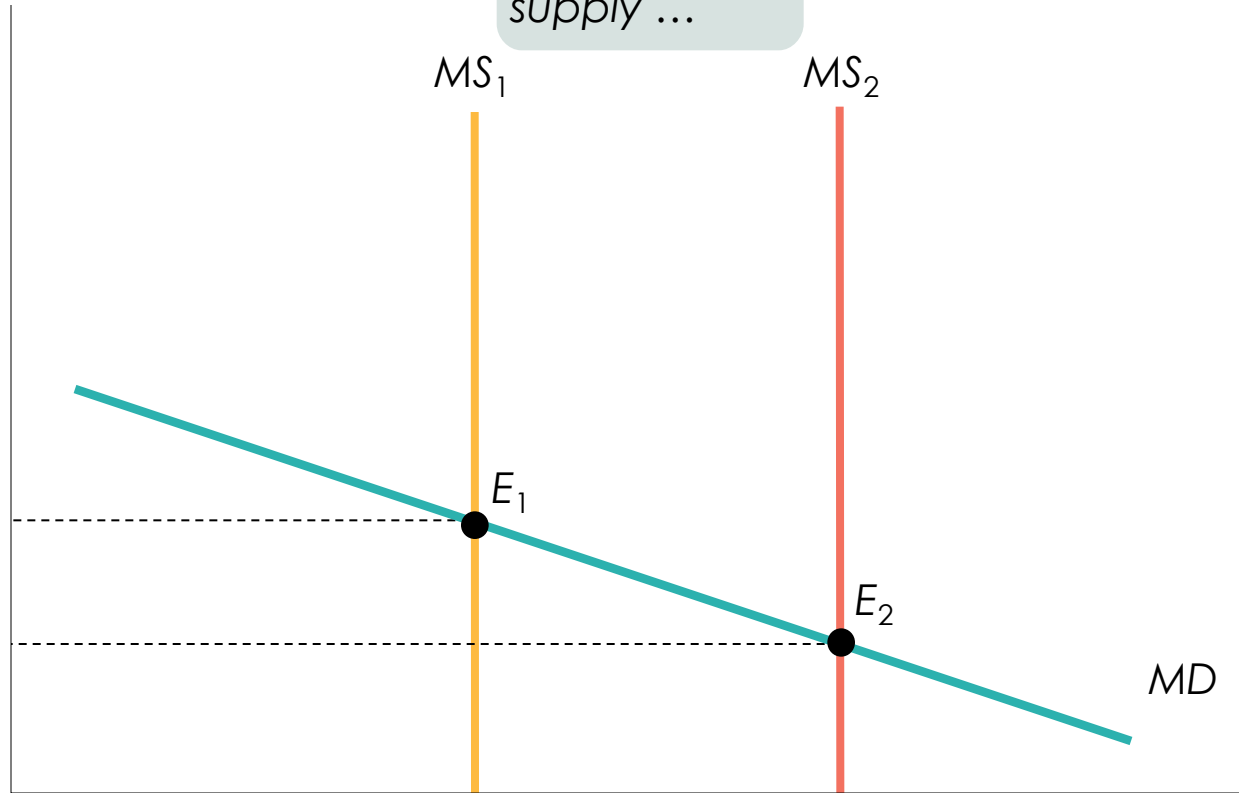
$\bar{M}_1$

$\bar{M}_2$

Quantity of money

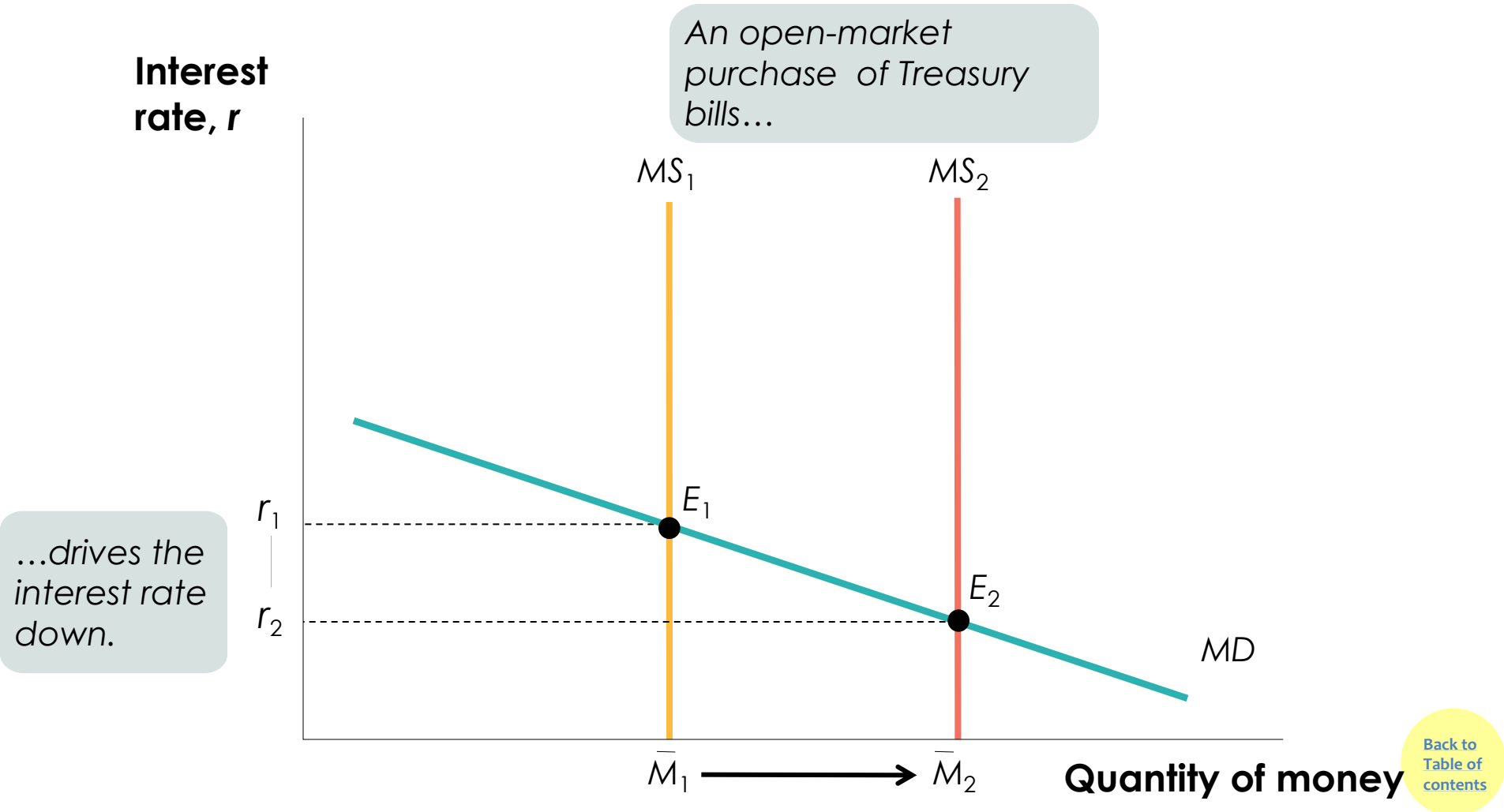
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...leads to  
a fall in the  
Interest  
rate.



# SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to push the interest rate down to the target rate.

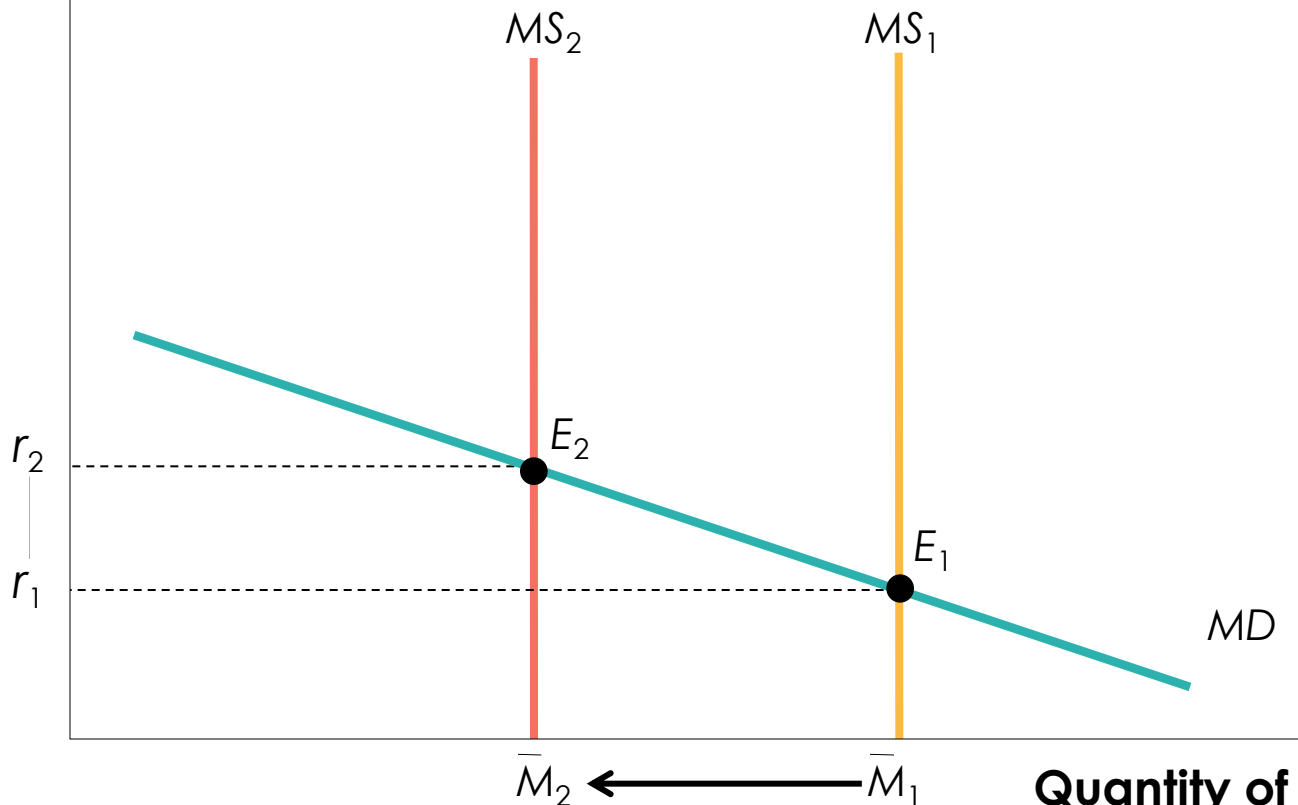


# SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to pull the interest rate up to the target rate.

Interest rate,  $r$

An open-market sale of Treasury bills...



...drives the interest rate up.

## LEARN BY DOING: PRACTICE QUESTION

**The money supply curve shifts to the left. The most likely cause for this shift is the FOMC's open-market:**

- a) purchase of Treasury bills.**
- b) sale of Treasury bills.**

# LONG-TERM INTEREST RATES

**Long-term interest rates don't necessarily move with short-term interest rates.**

**If investors expect short-term interest rates to rise, they may buy short-term bonds.**

*In practice, long-term interest rates reflect the market's average expectation for short-term rates.*



# MONETARY POLICY AND AGGREGATE DEMAND

***Expansionary monetary policy:*** monetary policy that increases aggregate demand (also called “easy money policy”).



***Contractionary monetary policy:*** monetary policy that reduces aggregate demand (also called “tight money policy”).

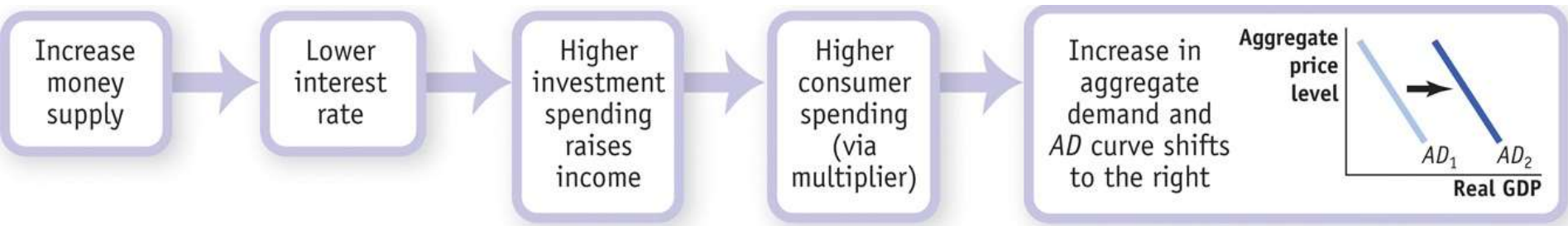
## LEARN BY DOING: PRACTICE QUESTION

**When the economy is producing at a short-run level of aggregate output that is less than potential output, it is most likely that the Fed will engage in open-market:**

- a) purchases of Treasury bills.**
- b) sales of Treasury bills.**

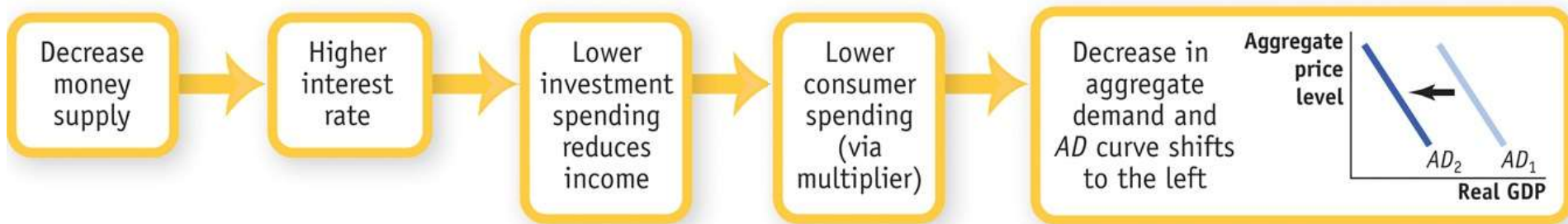
# MONETARY POLICY AND AGGREGATE DEMAND

## How the expansionary monetary policy cycle works:



# MONETARY POLICY AND AGGREGATE DEMAND

## How the contractionary monetary policy cycle works:



# MONETARY POLICY IN PRACTICE

**How does the Fed decide whether to use expansionary or contractionary monetary policy?**

**And how does it decide how much is enough?**

**Policy makers try to fight recessions, as well as to ensure *price stability*: low (though usually not zero) inflation.**

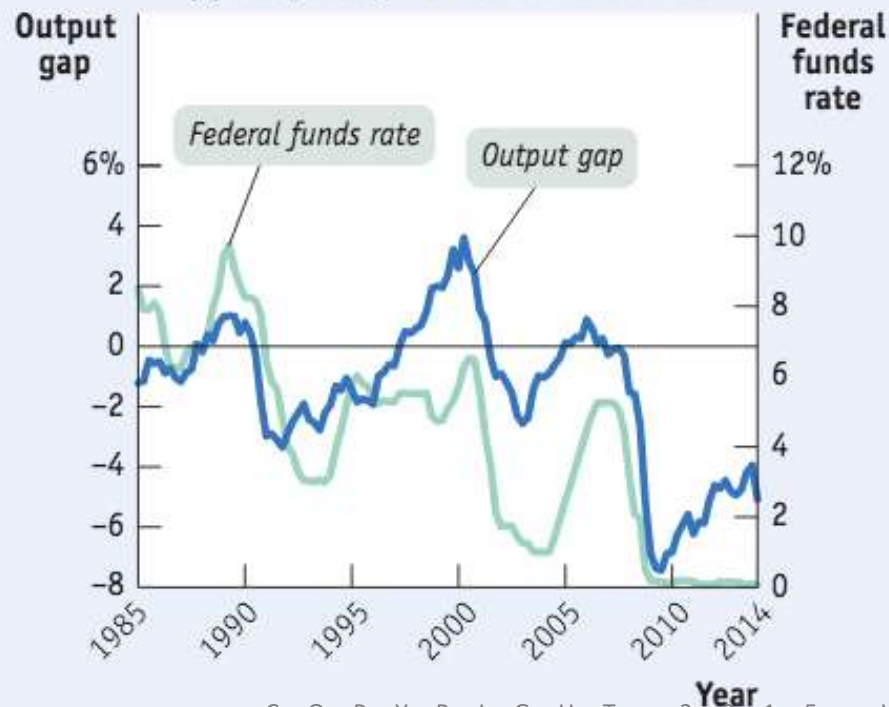
**Actual monetary policy reflects a combination of these goals.**

# TRACKING MONETARY POLICY

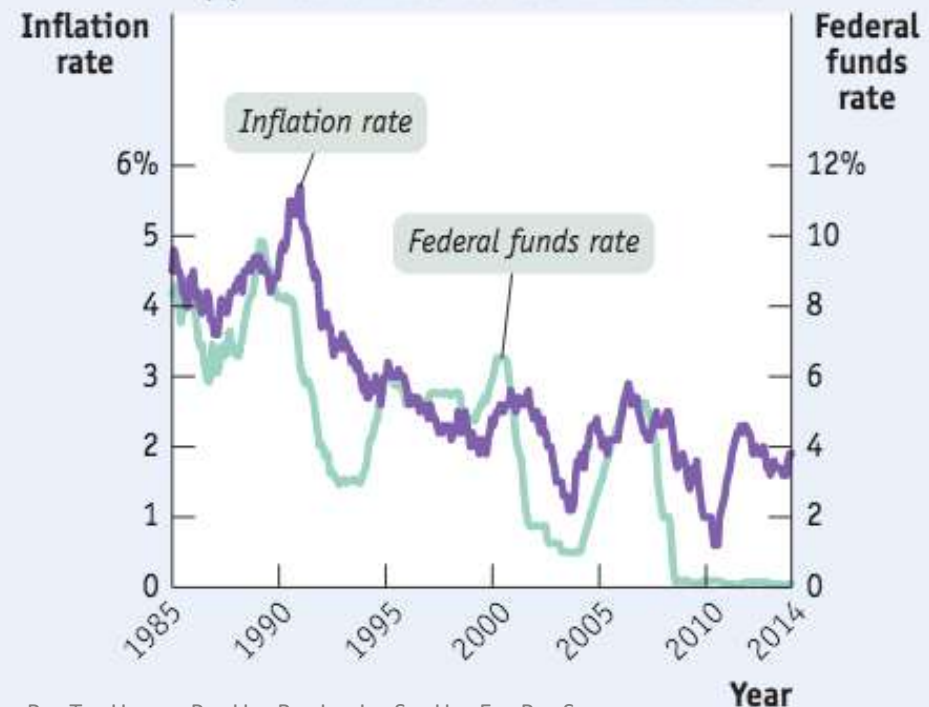
The federal funds rate usually **ris**es when the **output gap** is positive and **falls** when the **output gap** is negative.

The federal funds rate tends to be **high** when **inflation** is high and **low** when inflation is low.

(a) Output Gap vs. Federal Funds Rate



(b) Inflation Rate vs. Federal Funds Rate



# THE TAYLOR RULE FOR MONETARY POLICY

**In 1993, Stanford economist John Taylor proposed a simple rule for monetary policy.**

***Taylor rule for monetary policy:*** set the federal funds rate according to the level of the inflation rate and either the output gap or the unemployment rate.

# THE FEDERAL FUNDS TARGET AND THE TAYLOR RULE

**To address both goals:**

- inflation
- recessions

**The Fed can follow a rule:**



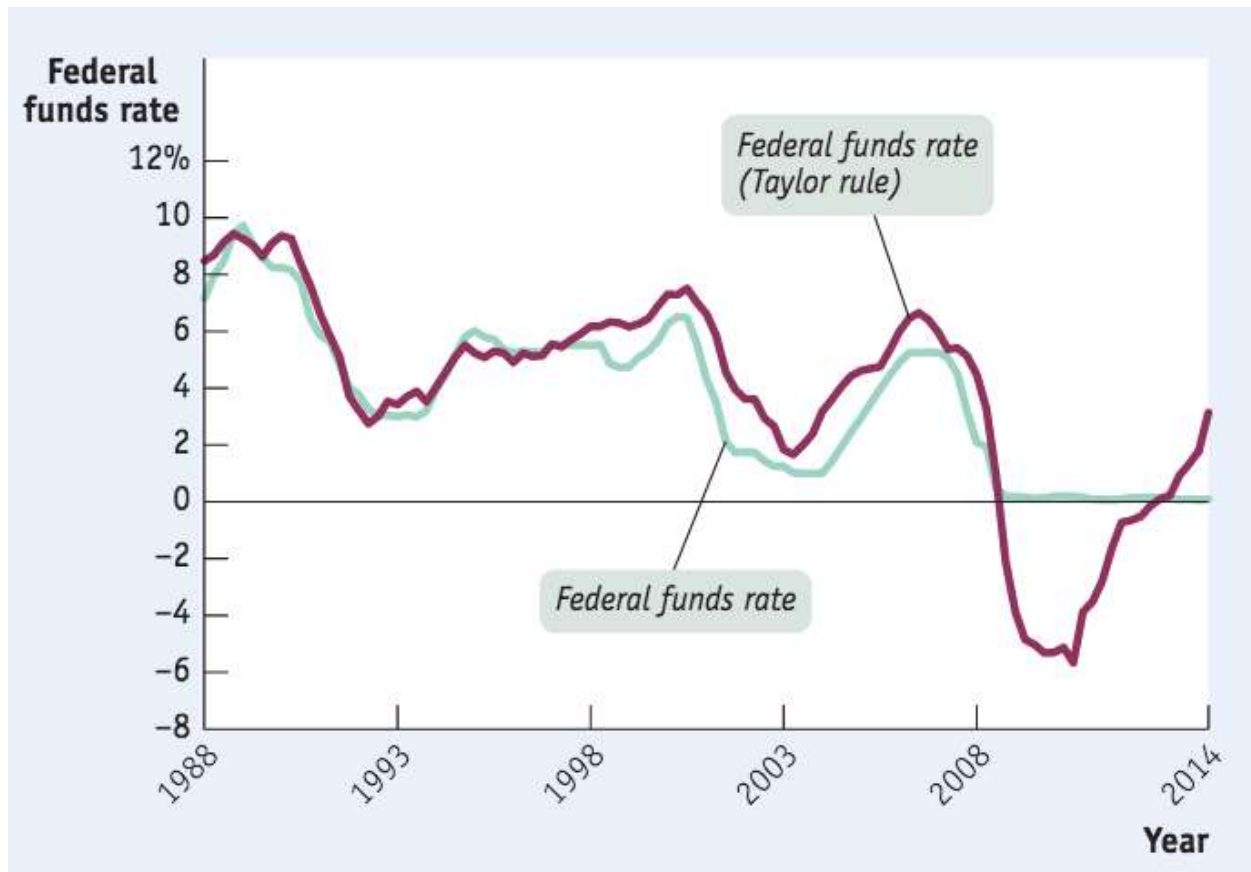
*Federal funds target rate =  
 $2.07 + 1.28 \times \text{inflation rate} - 1.95 \times \text{unemployment gap}.$*

**As it turns out, between 1988 and 2008 the Fed basically did just this.**

# TRACKING MONETARY POLICY

*The actual federal funds rate tracked the predicted rate quite closely through the end of 2008.*

*After that the Taylor rule called for negative interest rates, which aren't possible.*



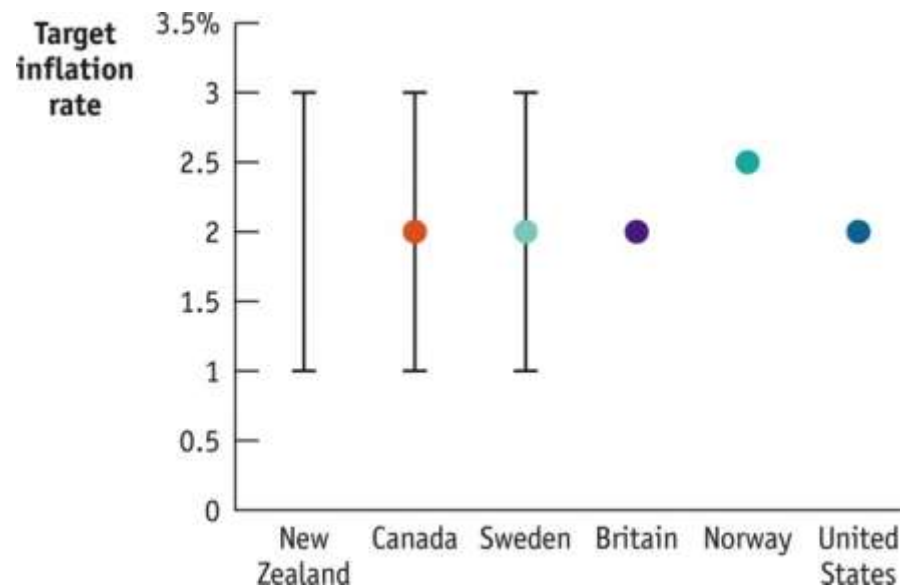
Sources: Bureau of Labor Statistics; Congressional Budget Office; Federal Reserve Bank of St. Louis; Glenn D. Rudebusch, "The Fed's Monetary Policy Response to the Current Crisis," FRBSF Economic Letter #2009-17 (May 22, 2009).

# INFLATION TARGETING



Instead of using a Taylor rule, many central banks use inflation targeting.

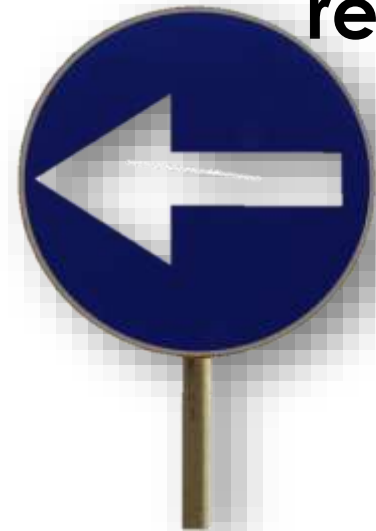
***Inflation targeting***: the central bank setting an **explicit target for the inflation rate** and using monetary policy to hit that target.



# DOES IT MATTER?

One major difference between the two methods:

- **Inflation targeting is forward-looking** (based on a forecast of future inflation).
- **Taylor rule is backward-looking** (adjusts monetary policy in response to *past* inflation).

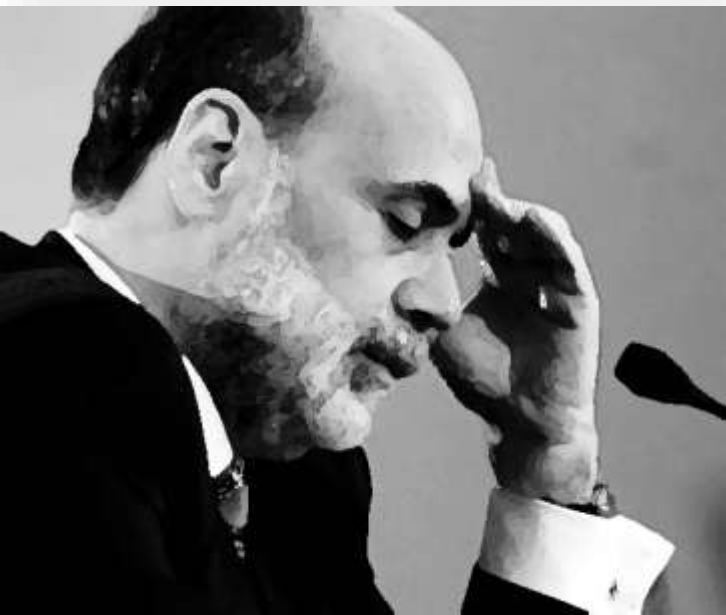


# THE ZERO LOWER BOUND PROBLEM

The fact that interest rates can't go below zero (“**zero lower bound**”) limits the power of monetary policy.

**November 2010: Interest rates were at or near zero, but the economy was still far below potential output.**

So the Fed got creative and tried “quantitative easing” (buying longer-term government bonds).



*The Fed hoped this would drive down long-term interest rates and have an expansionary effect on the economy.*

# MONEY, OUTPUT, AND PRICES IN THE LONG RUN

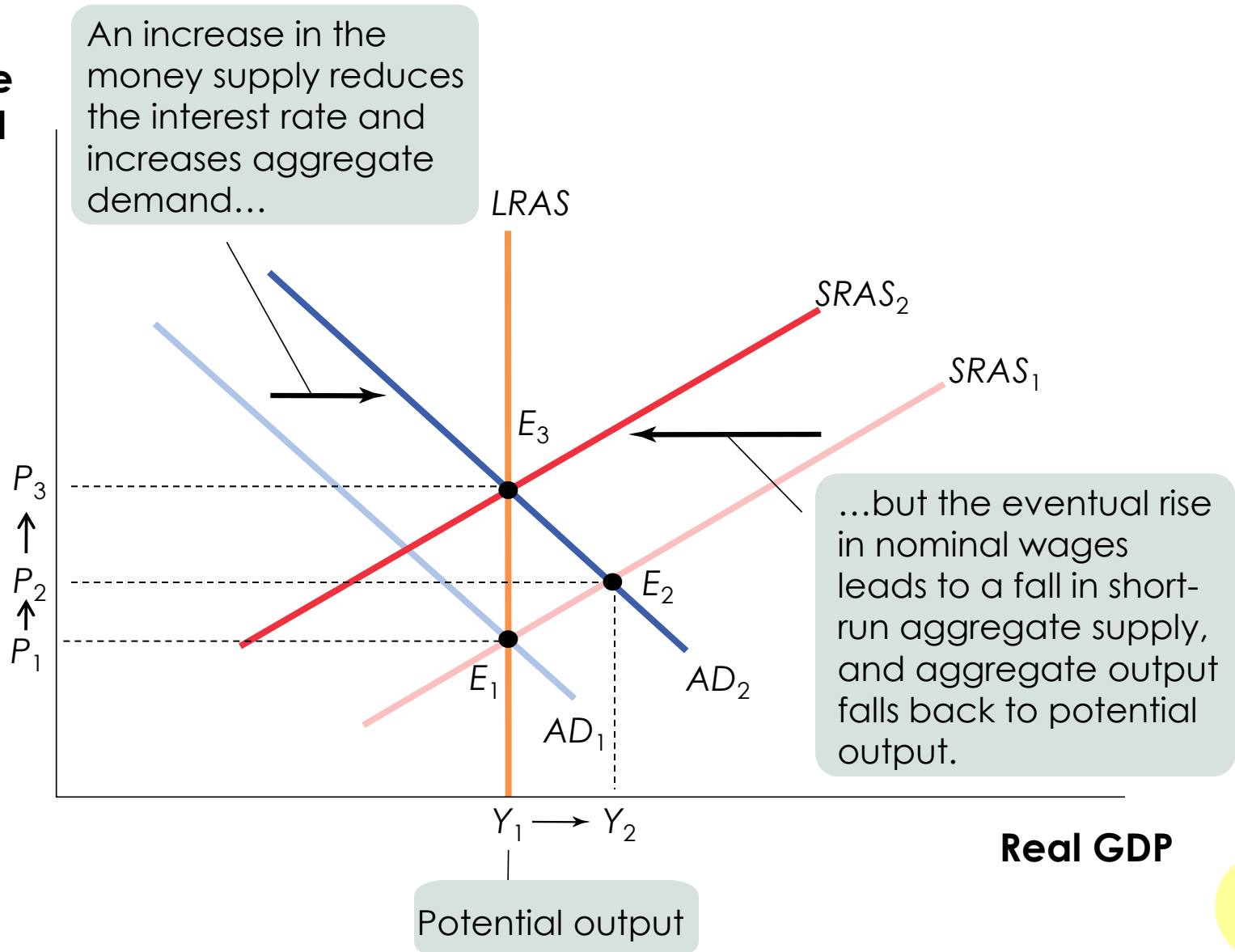
**What about the economy's ability to self-correct toward long-run equilibrium through flexible wages and *SRAS* shifts?**

**In short, we believe that in the long run, changes in the quantity of money affect the price level but not the output or interest rate.**



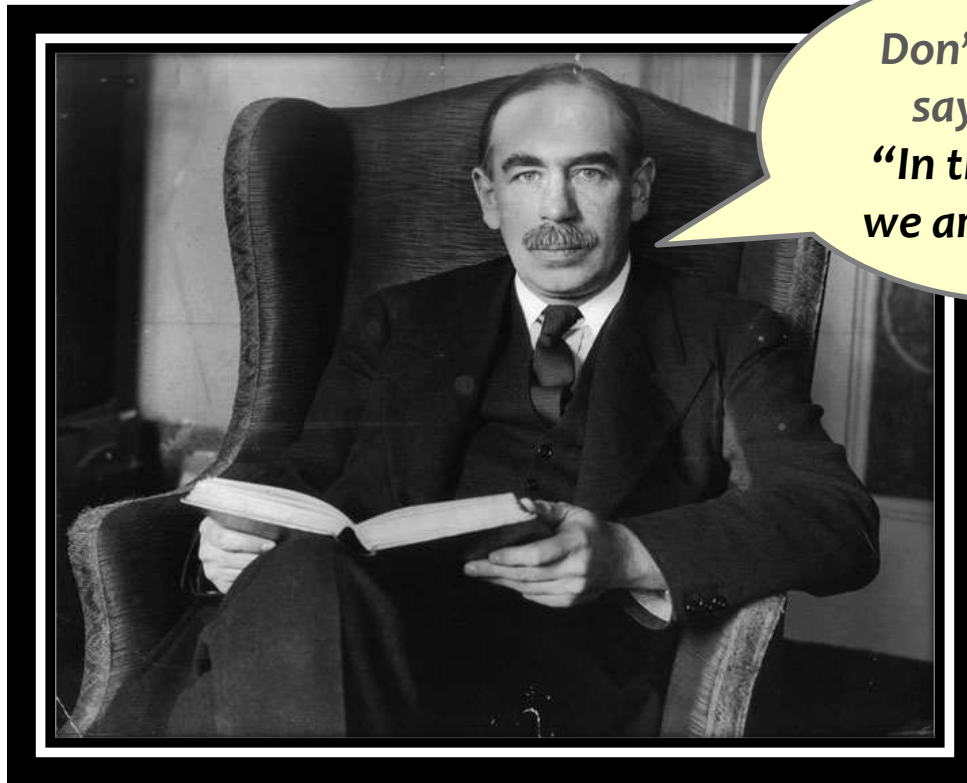
# MONEY, OUTPUT, AND PRICES IN THE LONG RUN

**Aggregate price level**



# MONETARY NEUTRALITY

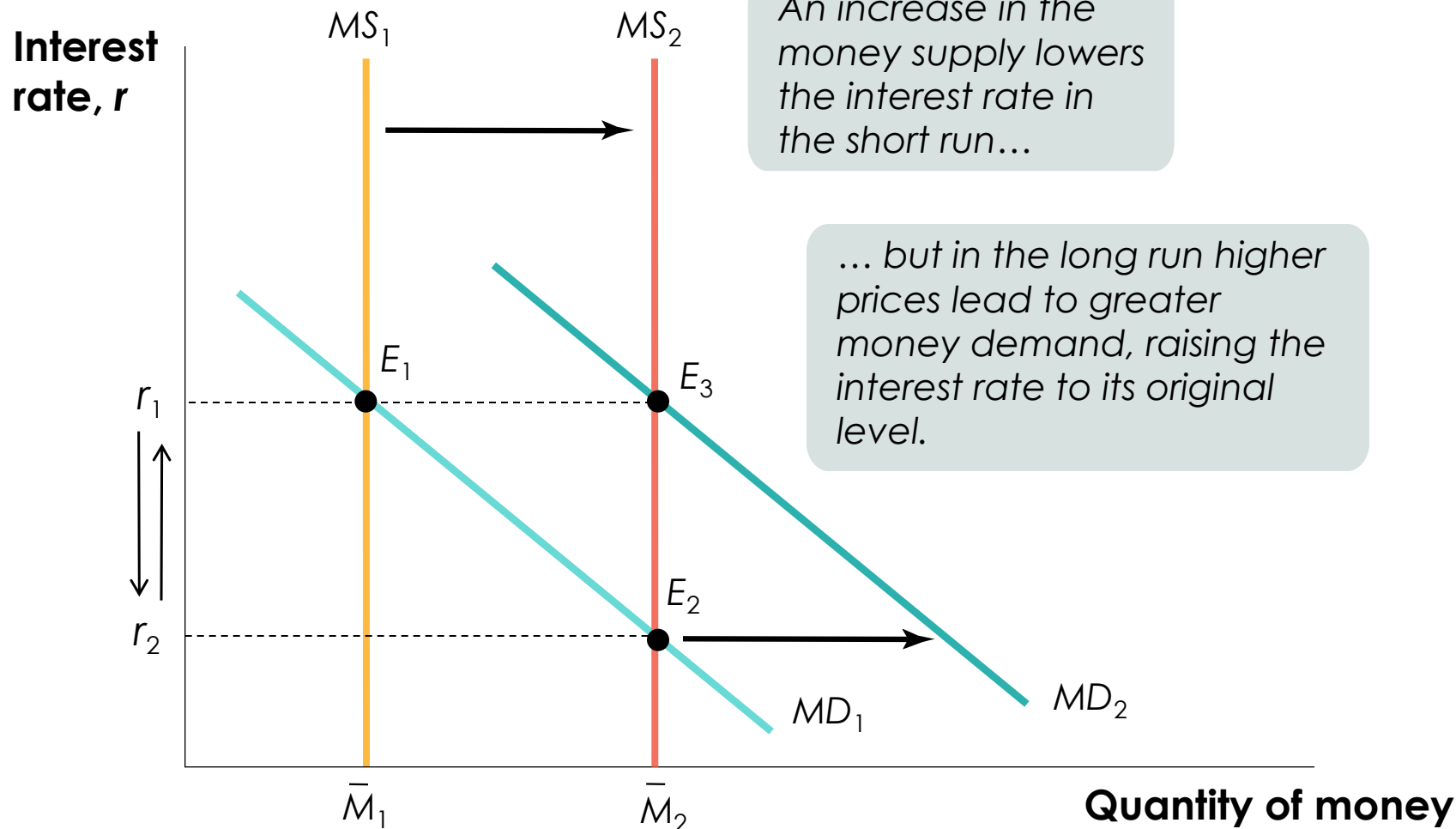
***Monetary neutrality:*** Changes in the money supply have no real effect on the economy.



Don't make me  
say it again:  
"In the long run  
we are all dead."

# LONG-RUN CHANGES IN THE MONEY SUPPLY AND INTEREST RATE

**In the long run, changes in the money supply don't affect the interest rate.**



# ECONOMICS IN ACTION

## INTERNATIONAL EVIDENCE OF MONETARY NEUTRALITY

Do increases in the money supply really lead to increases in the price level? It's not exact, but yes.

